EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Calling a C# function that returns a new balance automatically changes the caller's original int variable.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

DO NOW — RECALL

Programming paradigms: applied comparison

Name: _____ Class: _____ Date: _____

1. How does OOP differ from a procedural program?

2. What does a local variable mean?

3. If a field is private-set, how can callers change it safely?

4. Why does `Deposit(balance,5);` not update the caller's int balance if only a return value is produced?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

OOP groups state and behaviour into objects; procedural code is organised around procedures/functions.

2. Model answer

A variable accessible only within its declaration's scope.

3. Model answer

Through public methods/property logic defined by the class.

4. Model answer

The return is discarded and a value-type parameter is a copy; the caller must assign the returned result.

Vocabulary — know these terms

refactoring — Restructuring code while preserving expected behaviour.

encapsulation — Controlled access to internal state.

instance — Individual object created from a class.

return value — Output returned to the caller.

side effect — Observable change beyond a returned result.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare one maintenance advantage and one risk of procedural and OOP implementations for a library system.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Procedural AddFive(x) returns x+5. If caller x=2 discards the result, what is x?

Question 2

Two objects have balances A=10, B=30. A.Deposit(7) and B.Deposit(2) run. Find both balances and describe why they differ.

NEW LEARNING

Read, annotate and explain

Procedural state

The caller passes an integer balance and explicitly stores the returned value.

Object state

Account object A stores Balance internally and Deposit changes that object's state.

Separate instances

Account B stays at 20 when Account A changes to 15; the objects have independent state.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Two accounts start at 8 and 20; only first account deposits 4. Trace both.

Predict

Account a's Balance becomes 12.

Model response

Account b remains at 20 because its state is separate.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve